

Ranking sensitivity by method and article bundle

Global / system-level methods

Decision Trees	0.04 [2]	0.03 [3]		0.03 [4]		0.02 [2]	0.03 [2]
RuleFit	0.25* [2]	0.17 [3]		0.19 [4]		0.15* [2]	0.18* [2]
RuleSHAP	0.25* [2]	0.13 [3]		0.19 [4]		0.15* [2]	0.13 [2]
MechaRule	0.18 [2]	0.08 [2]		0.22* [3]		0.12 [2]	0.09 [2]
MechaRule stage 1	0.06 [2]	0.02 [2]		0.06 [3]		0.03 [2]	0.02 [2]
PDP	0.35* [1]	0.36* [3]		0.40* [3]		0.26* [1]	0.34* [1]
ICE	1.00* [1]	0.54* [2]		0.17 [1]		0.45* [2]	0.78* [1]
Global SHAP	0.10 [1]			0.16 [1]		0.05 [1]	0.03 [1]
	AIA 11 + Annex IV	AIA 13-14	DSA 17	DSA 27 + P2B 5	GDPR + AIA 86 + MiFID 25	MDR	MiFID 17

Local / decision-level methods

LIME		0.00 [2]	0.01 [1]		0.01 [1]	0.00 [1]	0.00 [2]
Anchors		0.01 [2]	0.53* [2]		0.53* [2]	0.01 [2]	0.01 [2]
CEM			0.53* [2]		0.53* [2]	0.05 [1]	0.03 [1]
DiCE		0.51* [2]	0.52* [2]		0.52* [2]	0.01 [1]	0.01 [1]
ProtoDash		0.22 [1]	1.00* [1]		1.00* [1]	0.32 [1]	0.22 [1]
Exact local SHAP			0.10 [1]		0.10 [1]	0.11 [1]	0.06 [1]
Approximate local SHAP			0.05 [1]		0.05 [1]	0.03 [1]	0.02 [1]
CAVs		0.00 [1]	0.03 [1]		0.03 [1]	0.01 [1]	0.00 [2]
DeepLIFT		0.05 [2]	0.11 [1]		0.10 [1]	0.04 [1]	0.05 [2]
DeepSHAP		0.08 [2]	0.15 [1]		0.13 [1]	0.08 [1]	0.08 [2]
LRP		0.01 [1]	0.09 [1]		0.05 [1]	0.01 [1]	0.01 [2]
Grad-CAM		0.02 [1]	0.09 [1]		0.13 [1]	0.02 [1]	0.03 [2]
Integrated Gradients		0.09 [1]	0.19* [1]		0.17* [1]	0.15* [1]	0.10 [2]
Attention		0.01 [1]	0.04 [1]		0.07 [1]	0.01 [1]	0.01 [2]
	AIA 11 + Annex IV	AIA 13-14	DSA 17	DSA 27 + P2B 5	GDPR + AIA 86 + MiFID 25	MDR	MiFID 17

* baseline soft-score winner for at least one question; [n] = eligible question cells; grey = ineligible

Mean probability of top rank under perturbation

1.0

0.8

0.6

0.4

0.2

0.0